

Lithium Niobate Piezoelectric Actuator-Integrated Fiber Fabry-Pérot Tunable Filter with Ultrahigh Speed and Linearity

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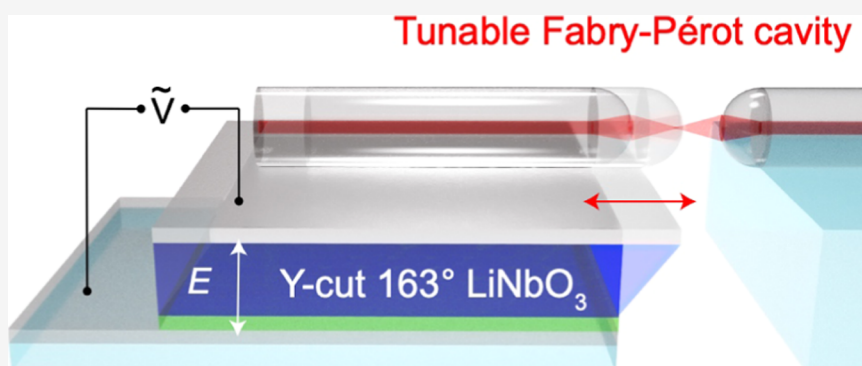
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ABSTRACT: Fast-tunable filters play a crucial role in various applications including tunable lasers, high-speed optical communications, microwave photonics, and spectroscopy. However, current fiber Fabry-Pérot tunable filters (FFPTF) predominantly rely on piezoelectric transducers (PZT), which struggle to achieve rapid modulation with high repeatability and accuracy owing to hysteresis and creep effects. In this study, we demonstrate an FFPTF with high linearity (>0.99) and an approximate 1 MHz dynamic tuning frequency in the resonant state using the thickness-shear resonance of a Y-cut 163° lithium niobate (LiNbO_3) sheet. The device exhibited a high overall performance with a finesse of 118 and an insertion loss of 2.54 dB, with well-designed fiber cavity configurations. This study provides a new approach for precise and rapid wavelength manipulation and selection, thereby paving the way for high-speed optical modulation and analyses.

KEYWORDS: lithium niobate, piezoelectric, Fabry-Pérot, fiber filter, tunable filter

INTRODUCTION

High-speed wavelength tuning is essential for optical fiber communication,^{1,2} microwave photonics,^{3–5} lasers,^{6–8} spectroscopy,^{9–11} sensors interrogation,¹² and many other fields, where tunable filters are key elements. The Fabry-Pérot tunable filter (FPTF) is a versatile and reliable modulation approach with a high resolution and low optical insertion loss.^{10,13} The mechanism of FPTF includes refractive index tuning, such as electro-optical¹⁴ and liquid-crystal tuning,^{15,16} as well as cavity-length tuning, such as mechanical, electrostatic,^{17,18} and piezoelectric tuning.^{19,20}

Among these, electro-optic tunable filters^{21–23} based on materials, such as lithium niobate (LiNbO_3), exhibit high-speed operation up to the gigahertz level. However, the relatively weak electro-optic coefficient and strong second-order Kerr effect of LiNbO_3 restrict the tuning range.²⁴ The electrically controlled birefringence of liquid crystals²⁵ typically exhibits an extensive tuning range, poor temperature stability, and slow response times of approximately 1 ms.²⁶ Electrostatic tuning is commonly used for FPTF in the form of microelectro-

mechanical systems,²⁷ suffering from the relatively poor linearity caused by the nonlinear relationship between the electrostatic force and displacement.²⁸ A fiber FPTF (FFPTF) based on piezoelectric tuning²⁹ offers a compact footprint, low loss, and large tuning range. However, current FFPTF is mainly realized using piezoelectric transducers (PZT), which can only be tuned at a rate of up to tens of kilohertz. The intrinsic creep and hysteresis effects of PZT are inevitable and limit the system accuracy and repeatability.^{30–32} In addition, PZT has a low Curie temperature^{33,34} and poor thermal stability, which limits its application. Therefore, these approaches have limitations in terms of the mechanisms and intrinsic properties of the materials. None of them can

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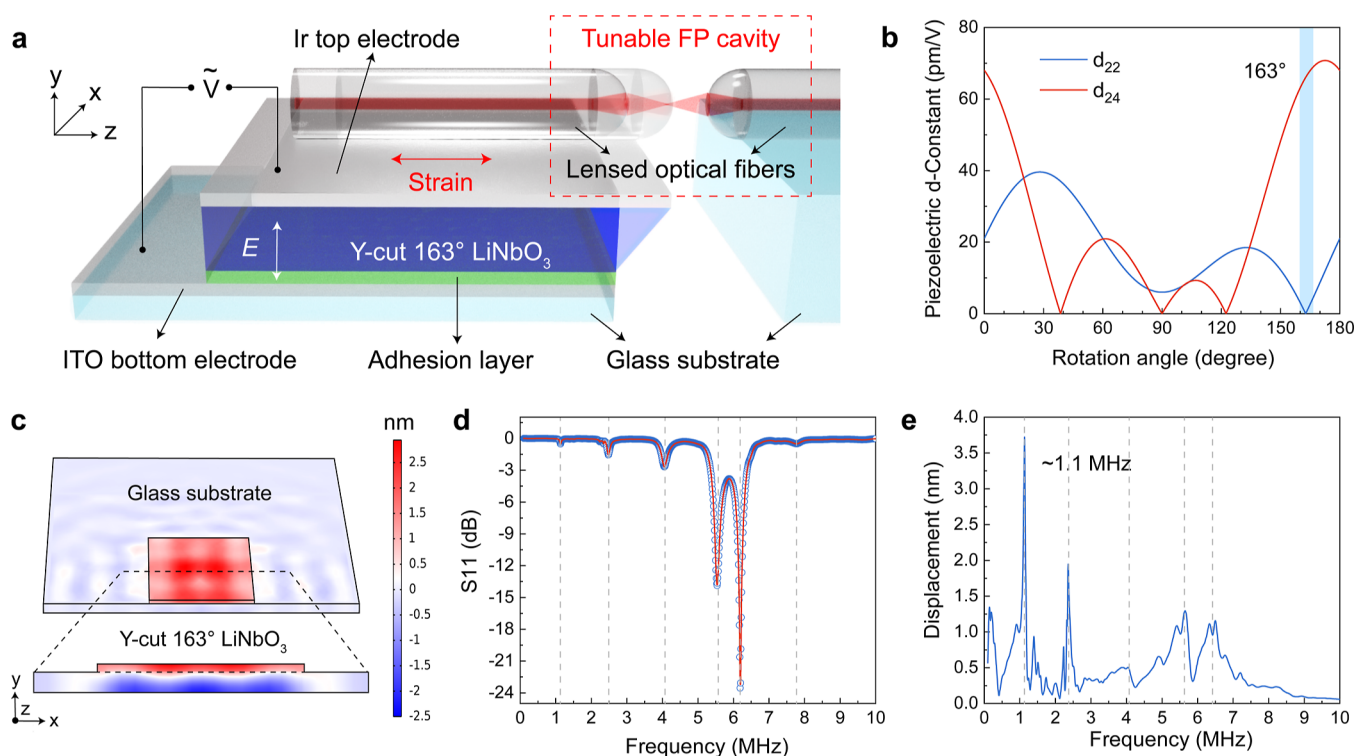


Figure 1. Design and simulation of the LiNbO₃ actuator. (a) Thickness–shear motion of Y-cut 163° LiNbO₃ in an electric field. (b) Piezoelectric strain constant of Y-cut LiNbO₃ versus rotation angle. (c) Top (upper panel) and cross-sectional (lower panel) views of the simulated displacement distribution in the *z*-direction of the LiNbO₃ actuator at a resonance frequency of 1.105 MHz. (d) Simulated spectral response of *S*₁₁ of the actuator (see the Methods section for details). (e) Simulated displacement of the actuator as a function of driving frequency.

simultaneously satisfy the requirements of these two main aspects: (i) excellent tuning characteristics, such as extensive range and high speed, and (ii) excellent system characteristics, such as low insertion loss, high linearity, repeatability, and stability.

LiNbO₃ is a promising material with excellent piezoelectric properties, such as a large electromechanical coupling coefficient, high Curie temperature, low hysteresis, and creep.^{35,36} Several studies have investigated LiNbO₃ piezoelectric devices, such as ultrasonic transducers^{37–39} and photoelastic modulators.⁴⁰ However, owing to difficulties in assembly and processing,^{41–45} LiNbO₃ has never been integrated as a displacement actuator in compact and integrated all-fiber devices.

In this study, we utilized the excellent piezoelectric properties of LiNbO₃ in a flat-plate actuator to achieve high speed and low hysteresis characteristics for tunable filters. We present a LiNbO₃-based FFPTF with a 0.5% nonlinear error and an approximate 5.6 nm tuning range under voltages from −100 to +100 V using the thickness–shear resonance of Y-cut 163° LiNbO₃. The FFPTF can operate in a resonant state with a tuning frequency of approximately 1 MHz. The FFPTF was constructed using a pair of LiNbO₃ sheets under two fibers integrated with microlens on the end face. It exhibited a high overall performance with a finesse of 118 and an insertion loss of 2.54 dB, using highly reflective films and ultraviolet (UV) gel microlens on oppositely positioned dual-fiber end faces. This study provides a new approach for precise and rapid wavelength manipulation and selection, paving the way for high-speed optical modulation and analyses.

RESULTS

Design and Simulation of the Actuator. The displacement of the piezoelectric actuator under an electric field is intrinsically linked to the piezoelectric constant.⁴⁶ Owing to its distinguished anisotropic crystalline structure, LiNbO₃ exhibits a notable correlation between its piezoelectric constant matrix *d*_{*ij*} and the inherent orientation of its crystallography.⁴⁷ Figure 1a depicts the thickness–shear motion of Y-cut 163° LiNbO₃ along the *z*-direction under a *y*-directional electric field. A Y-cut 163° LiNbO₃ sheet, 12 × 10 mm, was selected to fabricate the piezoelectric actuator, which enabled the length adjustment of the Fabry–Pérot (FP) cavity. The actuator structure comprises a 60 nm iridium electrode, a 350 μm LiNbO₃ layer, an approximately 10 μm paraffin layer, a 220 nm indium tin oxide (ITO) electrode, and an approximate 1.1 mm glass substrate, arranged from top to bottom (see the Methods section for details). The piezoelectric coefficients, *d*₂₂, and *d*₂₄, represent the thickness extension and thickness shear, respectively. For a Y-cut angle of 163° for LiNbO₃, *d*₂₄ reached its maximum, whereas *d*₂₂ became zero, as depicted in Figure 1b. This resulted in the observation of a predominantly thick shear strain *S*_{*yz*} in the *yz*-plane.

Simulations of the LiNbO₃ piezoelectric actuator were performed by using finite-element simulation. Figure 1c shows the displacement distribution in the *z*-direction of the actuator during resonance under a radio frequency (RF) signal at a frequency of 1.105 MHz. The cross-sectional view of the strain distribution revealed the resonance of the glass substrate with LiNbO₃ in this vibration mode (see Supporting Information for details). The *S*₁₁ curve of the actuator obtained through the simulation is presented in Figure 1d, and the variation in the displacement at a specific point on the actuator surface with

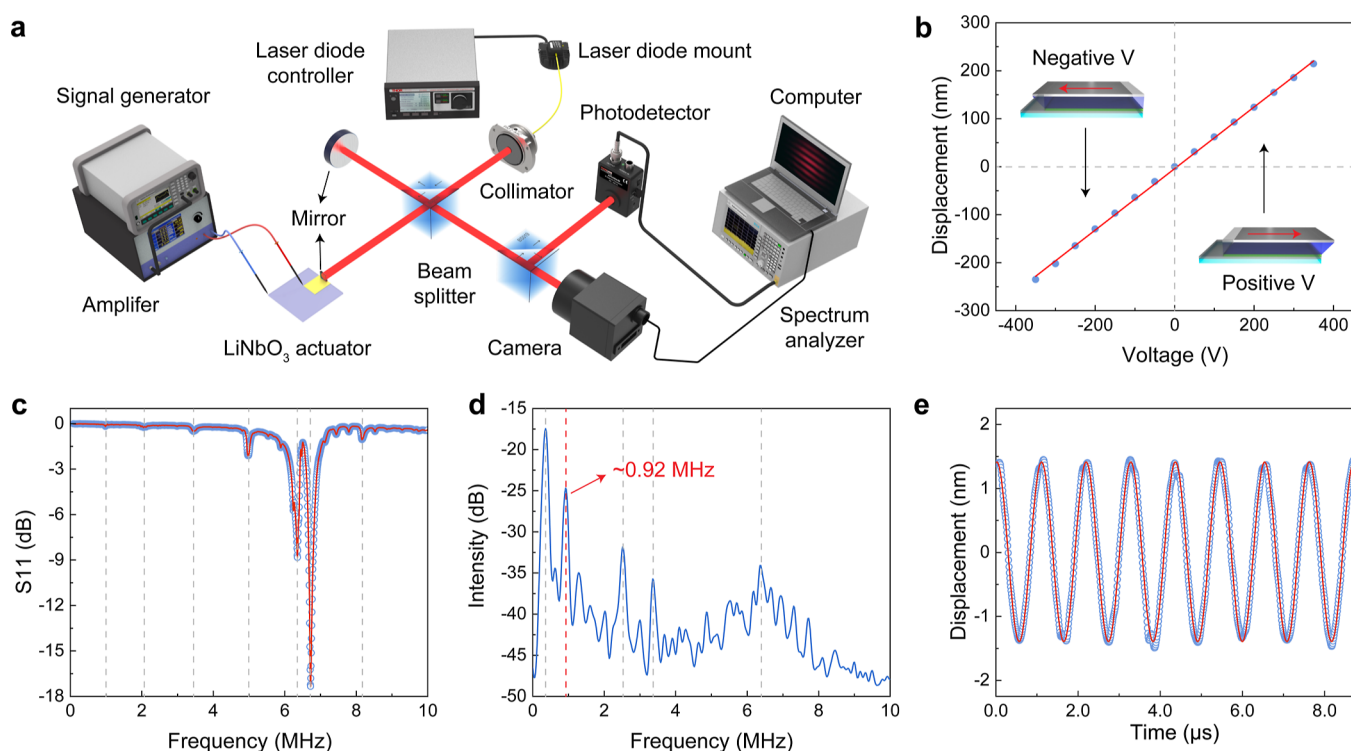


Figure 2. Measurement of the LiNbO₃ actuator. (a) Michelson interferometer for measuring the vibration displacement of LiNbO₃ actuators. (b) Displacement of the LiNbO₃ actuator as a function of different direct current (DC) voltages. (c) Spectral response of actuator vibrations at different frequencies obtained using a spectrometer. (d) S_{11} of the LiNbO₃ actuator was measured using a network analyzer. (e) Time response of the vibration displacement at a frequency of 0.92 MHz. The AC is 20 Vpp.

respect to the driving frequency is illustrated in Figure 1e. Distinct resonant modes are highlighted by the peaks observed at various positions along the frequency axis of the S_{11} curve. The resonant peak near 1.1 MHz is particularly noteworthy because the displacement amplitude is maximized near this frequency, as depicted in Figure 1e. The peak at approximately 6.2 MHz in the S_{11} curve exhibited the highest contrast, indicating a low impedance. However, the displacement at this frequency is relatively small, as shown in Figure 1e. This is attributed to the decreased electromechanical coupling coefficient K^2 , which decreases quadratically with respect to the mode order n .⁴⁸

Characterization Measurements of the Actuator.

Detailed methods for fabricating the LiNbO₃ actuator can be found in the Methods section. A Michelson interferometer was used to evaluate the performance of the LiNbO₃ actuator. As shown in Figure 2a, the LiNbO₃ actuator was positioned along the reference optical path of the interferometer. The displacement of the actuator in the length direction was obtained by varying the voltage, as shown in Figure 2b. The results demonstrated a low nonlinear error of approximately 0.5%, which was significantly smaller than that of the PZT actuator with various compensation strategies.⁴⁹ A static displacement tuning rate of approximately 0.6 nm/V. This is smaller than the theoretical displacement (see the Methods section for details) of the Y-cut 163° LiNbO₃ calculated using the formula. This may be because the displacement calculated by the formula is in a boundary-free state, whereas the LiNbO₃ measured in the experiment adhered to the glass substrate by using paraffin, which may have hindered the displacement output of LiNbO₃. In addition, the actual piezoelectric

coefficient, d_{24} , of the LiNbO₃ material used in the experiments may be lower than the theoretical value.

The illustration in the lower corner shows the shear motion of the actuator in opposite directions at positive and negative voltages. Using a signal generator, we scanned the signal frequency from 0 to 10 MHz while maintaining the signal amplitude at 20 Vpp. The resulting spectral response, measured using a spectrometer, is shown in Figure 2c. In the spectral response plot, the horizontal and vertical coordinates of each peak correspond to the frequency and intensity of the response, respectively. A response is observed near the 1 MHz frequency, corresponding to the position near the 1.1 MHz in Figure 1e. Notably, this frequency deviated slightly from the results obtained through the finite-element simulations discussed in the preparation section. This discrepancy can be attributed to the additional weight of the mirror, which results in a slight reduction in the resonant frequency.

The measured electrical reflection coefficient S_{11} of the actuator is shown in Figure 2c. Notably, the figure exhibits multiple responses at 1, 2.5, 3.4, 5, 6.3, and 6.7 MHz, which correspond well with the spectral response depicted in Figure 2d and the simulated S_{11} parameters illustrated in Figure 1d. Consistent with the simulation, the maximal contrast peak observed in the S_{11} parameter examination exhibited diminished intensity within the spectral response plot. For the signal generator, the output frequency was set to 0.92 MHz with an intensity of 20 Vpp. The vibration of the actuator at this frequency was measured by using a laser vibrometer, as depicted in Figure 2e. The magnitudes of the experimentally measured dynamic displacements are relatively small compared with the simulated values in Figure 1e. There are several possible reasons for this, including the dielectric loss in the

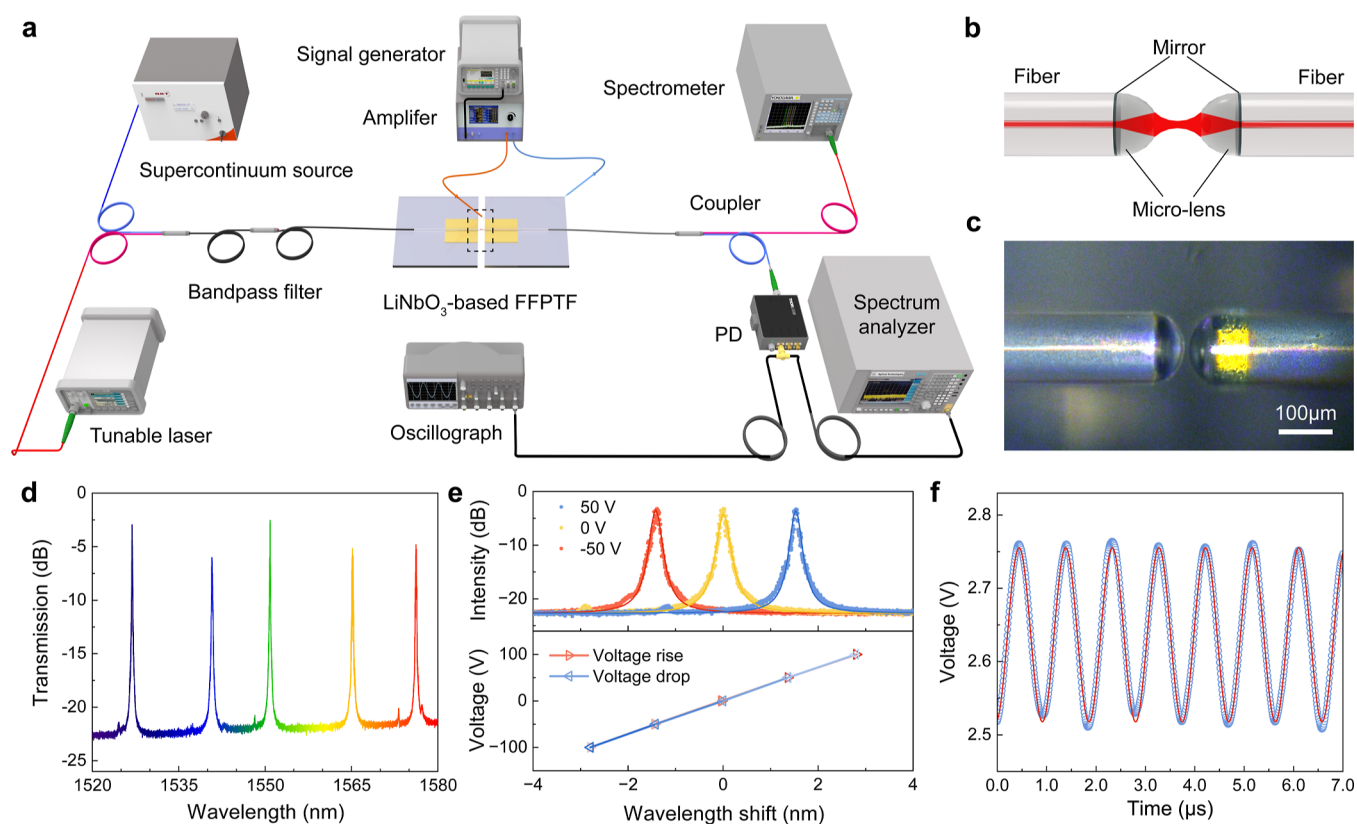


Figure 3. Characteristics of LiNbO₃-based FFPTF. (a) Schematic of the experimental setup for characterizing the LiNbO₃-based FFPTF. Schematic diagram (b) and microscope photograph (c) of the microlens in the FFPTF. (d) Measured transmission spectrum of FFPTF without voltage. (e) Transmission spectra (upper panel) of FFPTF with different DC voltages applied to the LiNbO₃ actuator. Hysteresis curves (lower panel) of LiNbO₃-based FFPTF. (f) Time response of the signal strength on the oscilloscope when the LiNbO₃-based actuator resonated at 1.06 MHz.

LiNbO₃ and substrate material,⁵⁰ mechanical damping introduced by the adhesive layer,⁵¹ and energy loss in the actuator due to airflow.⁵² These can be optimized by changing the materials used in future studies.

Characterization of the High-Speed Fiber FP Tunable Filter. Critical factors during the design process of FP filters include finesse, insertion loss, and free spectral range (FSR). To optimize the finesse and uphold the peak transmittance, we applied a dielectric film with an approximate reflectivity of 97.5% to the end face of the fiber. By fabricating a microlens on the fiber end face using UV glue (see the [Methods](#) section for details), the spot size of the resonant cavity mode was further reduced. This strategy effectively reduced diffraction and insertion losses, thereby improving the finesse of the filters.⁵³ We meticulously adjusted the fiber to the optimal position using a five-degree-of-freedom adjustment bracket and secured it to the LiNbO₃ actuator using a UV adhesive. [Figure 3b](#) shows a model diagram of the fiber FP filter with microlenses, and [Figure 3c](#) shows a photograph of one of the samples.

The experimental setup for both static and dynamic characterizations of the filter is shown in [Figure 3a](#). The filter measured here incorporates two microlenses, each with a thickness of approximately 28 μm, a radius of approximately 84 μm, and a refractive index of approximately 1.46. The distance between the two lens vertices was approximately 10 μm. [Figure 3d](#) shows the resulting transmittance spectra from which the FSR was estimated to be approximately 13 nm, corresponding to an effective cavity length of approximately 92 μm. The

transmittance spectra are obtained utilizing the spectrometer and supercontinuum light source. A 3 dB bandwidth of approximately 0.11 nm was formed around the center wavelength of 1550 nm, resulting in a finesse value of approximately 118. The quality factor is about 14,000, which is larger compare to other piezo-optomechanically tuning solutions.⁵⁴ However, the finesse of our device is still smaller than that of the strategy with concave cavity mirrors. For instance, the finesse of high-reflectivity concave cavity mirrors fabricated using the chemical etching process has reached the order of magnitude of millions.^{55–58} The intracavity losses of our device have not yet reached an adequately low level, and the reflectivity of our cavity mirror is still insufficient. These factors may contribute to the inadequate finesse achieved by our device. We can further improve the finesse of our device by utilizing concave cavity mirrors and improving the reflectivity of mirrors. The peak transmittance of the transmission spectrum indicates an overall insertion loss of 2.54 dB for the FP cavity. This relatively low loss can be attributed to the contributions of the highly reflective film and UV glue lens, which effectively suppressed the internal and coupling losses.

The passband width of the fiber bandpass filter between the light source and FFPTF was approximately 7 nm, which was used to filter signals outside the device signal channel range. The upper panel of [Figure 3e](#) shows the transmission spectra under voltages of −50, 0, and 50 V around a wavelength of 1550 nm. The transmission spectrum is obtained utilizing the spectrometer and supercontinuum light source and undergoes a blueshift of 2.9 nm as the voltage increases from −50 to 50 V.

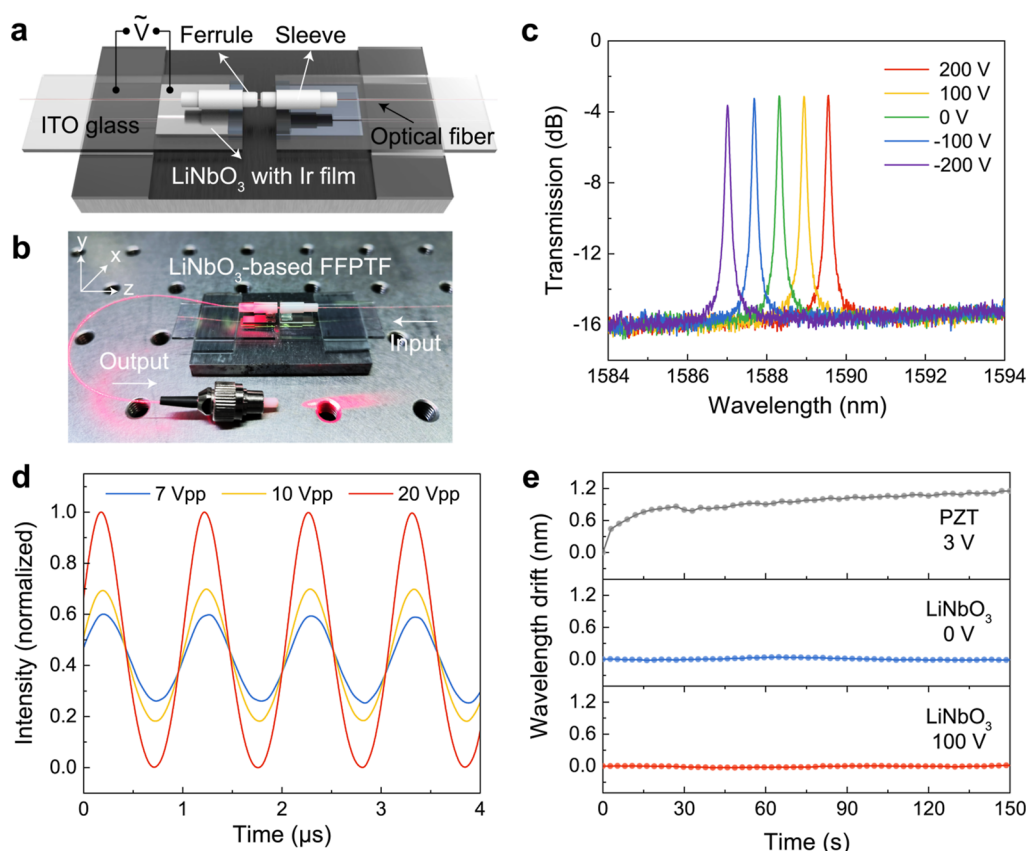


Figure 4. Characteristics and testing of the assembly sample. (a) Schematic of the structure of the assembled sample. (b) Photograph of the assembled sample. (c) Transmission spectra of FFPTF when different DC voltages were applied to the assembly sample. (d) Time response of the signal strength on the oscilloscope for different voltage amplitudes at an AC drive frequency of 0.96 MHz. (e) Creep testing of a PZT-based filter and assembled LiNbO₃-based FFPTF.

Because the cavity length of this filter is approximately 92 μm , we can deduce that the variation in the cavity length is approximately 175 nm, indicating a displacement tuning rate of 1.75 nm/V. This is still relatively small compared with the theoretical displacement tuning rate of 2.27 nm/V calculated in the [Methods](#) section. The difference in the displacement tuning rate between the different LiNbO₃ actuator samples may be due to variations in the stress and damping generated by the paraffin layer.

Notably, the insertion loss of the sample remained nearly constant during the tuning process, indicating the good stability of the filter. The hysteresis loop of the FFPTF is depicted in the lower panel of [Figure 3e](#), where the voltage-up and voltage-down response curves exhibit nonlinear errors as low as 0.5% and remarkable congruence. This is attributed to the minimal hysteresis and high linearity of the LiNbO₃ actuator, which are significant factors that affect the repeatability and positioning accuracy of the tunable filter. A tunable laser source was explored as the incident-light generator. The resulting optical output signal from the FFPTF under an alternating current (AC) was subsequently coupled to a photodetector. The output frequency of the signal generator was systematically swept at approximately 1 MHz under a peak-to-peak voltage of 74 V, and the amplitude fluctuations of the output signal from the photodetector were concurrently monitored using an oscilloscope. The frequency corresponding to the peak signal amplitude, as shown in [Figure 3f](#), was determined to be approximately 1.06 MHz.

Assembling and Packaging for Portable FFPTF. The samples used in previous tests were unassembled filters. However, the integration, stability, and portability of filters are important for their applications.⁵⁹ High-finesse FFPTF has extremely high requirements for the assembly of structures. It is necessary to ensure the smooth movement of the piezoelectric actuator and the alignment accuracy of the fiber end face. Therefore, the structure of FFPTF is highly precise and difficult to assemble. To improve integration, we assembled and improved the aforementioned sample, as shown in [Figure 4a,b](#). This filter had a finesse of approximately 99 and 3 dB bandwidth of about 0.1025 nm. The FSR is approximately 10.15 nm, corresponding to an effective cavity length of around 124.4 μm .

We also examined the tuning characteristics of the assembled samples by using the device diagram shown in [Figure 3a](#). The transmission spectrum, as depicted in [Figure 4c](#), manifests an approximate blueshift of 2.54 nm while the voltage changes from −200 to 200 V. We can deduce that the variation in the cavity length of this sample is approximately 200 nm, indicating a displacement tuning rate of approximately 0.5 nm/V. It was reduced compared to that of the unassembled sample, primarily because of the prestress induced during assembly and the stiffness of the assembled structures, which would result in a loss of output displacement.^{60–62} The diminished size of the piezoelectric sheet used in the assembled sample, measuring 6 mm wide and 10 mm long, may lower the mechanical stiffness of the actuator and lead to a decrease in the displacement. The insertion loss varies mildly

from 3.13 to 3.25 dB as the voltage increases from -100 to 100 V, indicating minimal fluctuations in the peak transmittance. We performed a frequency scan of the signal generator at approximately 1 MHz, while concurrently monitoring the amplitude variations on the oscilloscope. The frequency corresponding to the maximum signal amplitude displayed on the oscilloscope was determined to be approximately 0.96 MHz, as shown in Figure 4d. Changes in the amplitude of the driving voltage result in linear changes in the oscilloscopic response. This frequency slightly deviates from the 1.06 MHz observed in the unassembled sample. This deviation can be attributed to the additional weight of the ferrule in the overall FFPTF system, which causes a minor reduction in the resonant frequency.

To further investigate the creep behavior of the assembled sample, we set the applied voltage on the LiNbO_3 actuator to 100 V and recorded the variation in the transmission spectrum every three s for 150 s. The resulting wavelength–drift curve of the assembled sample plotted against the temporal axis is shown in Figure 4e. No significant drift occurred after the voltage was reset to zero or hundred volts. For comparison with a commercial product, we tested a PZT-based tunable filter. Figure 4e shows the creep behavior of this device. A noticeable creep effect was evident compared to the LiNbO_3 -based FFPTF, with a wavelength drift of approximately 1.15 nm in 150 s, whereas the LiNbO_3 -based FFPTF drifted by only approximately 0.04 nm. This is due to the electrodeposition in LiNbO_3 having a small lag compared to PZT when an external electric field is applied. A relatively low wavelength drift is significant for the stability of the tunable filter and affects its accuracy.

CONCLUSIONS AND DISCUSSION

Notably, the tuning range can be further improved. The reduction in the displacement observed in the assembled sample may be attributed to the prestress induced in the assembly and the decrease in the actuator surface area, which can be solved by increasing the size of the piezoelectric sheet. In addition, the thickness of the microlenses on the fiber end faces can simultaneously affect the finesse of the cavity and wavelength-tuning range of the filter (see Supporting Information Section III for details). Therefore, the thickness of the microlens should be further reduced while minimizing the decrease in the finesse. As we discussed in the results section, the stress introduced by the paraffin layer causes a decrease of the measured displacement tuning rate compared to a theoretical value of 2.27 nm/V and leads to differences in tuning rates between samples. Additionally, the paraffin layer may affect the resonance strength of the actuator under high-frequency driving. Considering these results, we would utilize the bonding process during the manufacturing of the LiNbO_3 -based FFPTF, which will improve the consistency and stability.

The simulation in Figure 1c reveals that LiNbO_3 resonates with the substrate, which complicates the situation but provides more freedom in the design of the actuator. For higher tuning speed, we could optimize the size and thickness of the glass substrate to increase the resonance frequency. Alternatively, structures can be fabricated on glass substrates to improve resonance strength. The first peak in Figure 2d appears to have a greater resonant strength. However, it occurred at a lower frequency of approximately 0.36 MHz. With proper optimization, it is possible to increase the resonant frequency to tens of MHz and present new

opportunities for ultrahigh-speed optical modulation and analysis.

In conclusion, we presented a high-speed FFPTF based on a LiNbO_3 actuator that exhibited remarkable characteristics including low hysteresis and creep. The FFPTF demonstrated excellent performance with a tuning speed of approximately 1 MHz, an insertion loss as low as 2.54 dB, and a finesse of up to 118 . By showcasing the enhanced tuning speed, repeatability, and accuracy, our work paves the way for precise and rapid wavelength manipulation and selection using ultrafast spectrometers, interrogators, tunable lasers, and microwave filters.

METHODS

Actuator Simulation and Calculation. The finite-element simulation was used to simulate the LiNbO_3 actuator. The simulation process used an electrostatic module, solid mechanics module, piezoelectric effect multiphysics field, and a frequency–domain study. First, three geometries of the LiNbO_3 , paraffin, and glass substrates were constructed from top to bottom of the geometric assembly. Their thicknesses were 350 , 20 , and 1100 μm , respectively, according to the real dimensions. Paraffin, as a mixture, has a density of approximately 840 kg/m^3 ; therefore, we used the modified polydimethylsiloxane material and modified its Young's modulus and Poisson's ratio to appropriate values. In the solid mechanics module, we set the glass and paraffin layers as linear elastic materials and added appropriate damping to restore the loss of real vibration. Based on the clamping method used in the test, the glass substrate was fixed as a line on the upper and lower surfaces. In the electrostatic module, we set the upper surfaces of LiNbO_3 and the glass substrate as the terminal and ground, respectively, and set the voltage of the terminal to 10 V. In the last step, a point probe was set at the position of the edge where LiNbO_3 and glass were tangential to each other, which was used to record the displacement that occurred at this point during the vibration process. In the frequency–domain study, the frequency sweep was set to a range of 0.1 – 10 MHz with a step size of 0.05 MHz. Following these calculations, a probe plot was constructed, as shown in Figure 1e. A global calculation was performed to obtain the S11 curve, as shown in Figure 1d. Add the body plot in the 3D plotting group and set the expression to “w” to get the displacement distribution of the actuator in the z-axis direction as shown in Figure 1c.

The strain of a Y-cut 163° LiNbO_3 sheet under an electric field in the thickness direction can be determined using the equation $S_{yz} = d_{24} \cdot E_y$. The specific thickness–shear displacement that occurs is expressed as $\Delta L = S_{yz} \cdot L$. Where L is the length, and the piezoelectric constant d_{24} of Y-cut 163° LiNbO_3 is approximately 66.3×10^{-12} m/V. The electric field strength E_y is equal to U/d , where U is the voltage, and d is the thickness of the actuator. At a voltage, U of 100 V and thickness d of 350 μm , the thickness–shear displacement can be calculated to be approximately 227 nm. The theoretical displacement tuning rate reached 2.27 nm/V.

FFPTF Fabrication. The production of FFPTF is mainly divided into three parts: the production of actuators, the production of fiber end polymer microlenses, and the overall assembly. We first used Y-cut 163° LiNbO_3 with a thickness of approximately 350 μm and a size of four inches to fabricate a piezoelectric actuator. LiNbO_3 wafers of this size specification can be purchased directly, and the LiNbO_3 wafer can be divided into small pieces with a width of 10 mm and a length

of 12 mm by using a scribing machine. An electron-beam evaporation device (Kurt J. Lesker, PVD75) was used to deposit 30 nm thick chromium and 100 nm thick gold on the surface of LiNbO₃ as the upper electrode of the actuator. Fiber end microlenses were fabricated using imaging microscopes and electric displacement tables. The end face of the fiber coated with the high-reflectivity film was fixed vertically downward on the fixture, and an electric displacement table was used to move it slowly until it encountered the UV glue (LEAFTOP 9310) droplet directly below. The UV glue has a refractive index of approximately 1.46. The height of the formed lens can be controlled by controlling the depth of fiber insertion. After standing for a period of time, the shapes were exposed to UV light.

After the LiNbO₃ actuator and fiber FP cavity were prepared, the fiber was meticulously adjusted to the optimal position using a five-degree-of-freedom adjustment bracket. The purchased glass was coated with an approximately 220 nm thick ITO film as the substrate with a length of 30 mm, width of 15 mm, and thickness of approximately 1.1 mm. The prepared electrode-coated LiNbO₃ plate was fixed onto an ITO glass substrate using paraffin, and the ITO glass was then bonded to a stainless-steel base using paraffin to enhance the stability of the device. UV glue was used to bond a small piece of strip-shaped glass slide to two symmetrically placed LiNbO₃ slices. The two ends of a cannula with a length of approximately 10 mm and an inner diameter of approximately 2.5 mm were bonded to a glass slide with UV glue. Next, a saw blade was used to cut the sleeve in half in the middle. Two ceramic ferrules with outer diameters of approximately 2.49 mm. Two optical fibers with ends coated with a high-reflection film and microlens are inserted into the two ferrules. Finally, the ferrule was inserted into the sleeve and the distance between the two optical fibers was adjusted to an appropriate size using a microstage. Two fibers were then sealed and fixed by using UV glue.

Experimental Setup for Actuator Characterization.

For the RF characterization of the modulator, we performed an S₁₁ scan using a network analyzer (Agilent E5072A) with an excitation power of 1 dBm and a bandwidth of 10 kHz for a broad scan from 0 to 10 MHz. The two leads of the coaxial line were connected to the upper and lower surface electrodes of the actuator; the measured reflection coefficients are shown in Figure 2d.

To characterize the vibration of the modulator, we constructed a Michelson interferometer structure using a laser diode light source (Thorlabs LP660-SF50), a visible photodetector [Thorlabs APD130A(/M)], and a camera. We bonded a small piece of the reflector to a LiNbO₃ sheet to measure the display of the actor in the length direction. The LiNbO₃ actuator modulates the interference fringes by changing the optical path difference through piezoelectric displacement in the interferometer optical path. The interfering signals were fed to the photodetector and camera through a beam splitter. The DC voltage generated by the signal generator (Agilent 33522A) was amplified using a high-voltage amplifier and subsequently applied to the upper and lower surface electrodes of LiNbO₃, creating a stable electric field. The voltage amplitude was varied and the positional movement of the interference fringes was recorded using a camera. The phase change and optical range difference can be determined by analyzing the pixel count that exhibits fringe movements. The AC sinusoidal signal generated by the signal

generator was loaded onto the actuator, and a visible photodetector was used to convert the jumping interference fringes into a current signal with sinusoidal amplitude variation. The signal was then transmitted to the spectrometer via a coaxial cable.

Experimental Setup for FFPTF Characterization. We used a supercontinuum light source (NKT Photonics SuperK Ursa) and a spectrometer (YOKOGAWA AQ6370) for static characterization. To assess the static tuning characteristics of the FFPTF, we applied different voltages to the LiNbO₃ actuator and observed a shift in the transmission spectrum. To evaluate the linearity and hysteresis effects of the FFPTF under static voltage tuning, we systematically varied the voltage on the LiNbO₃ actuator in 50 V increments, covering a range from −100 to +100 V. Concurrently, we recorded the corresponding center wavelength values and plotted them to generate the hysteresis loop of the FFPTF. To assess the dynamic tuning behavior of the high-speed tunable filter under AC conditions, we employed a sinusoidal AC waveform generated using a signal generator. Subsequently, the signal was used to impart a driving voltage of 20 V_{pp} to the FFPTF. We employed a tunable laser source (Agilent 81980A) as the incident-light generator and carefully aligned its central wavelength in proximity to the transmission peak of the FFPTF. The signal was transformed into an electrical waveform using a photodetector (Thorlabs PDB 430C), and subsequently visualized using a signal oscilloscope (Keysight MSOX6004A).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsphotonics.3c01740>.

Simulation analysis of actuators; calculation of microlenses with optical design software; effect of lens thickness on the tuning range of the FFPTF; and eye diagram measurement for FFPTF (PDF)

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Author Contributions

▽H.T. and Y.X. contributed equally. All authors provided active and valuable feedback on the manuscript. F.X. and H.T. initiated the concept and designed the studies; F.X. supervised the work; H.T. led the experiments and collected the overall data; F.X., H.T., and Y.X. co-wrote the paper; Y.X. helped draw figures and revise the paper; H.D. helped the actuator test and drew figures; Z.C. and Z.Q. helped the simulation. W.C. and S.Y. advised on the experiment. J.T. advised on the manuscript.

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Notes

The authors declare no competing financial interest.

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